NeuroReach

EEG-Assisted Stroke Rehabilitation

A Wellness and Engagement Adjunct | krishna-bhatt.com



Krishna Bhatt

The Problem

Stroke rehab has a therapy-intensity gap

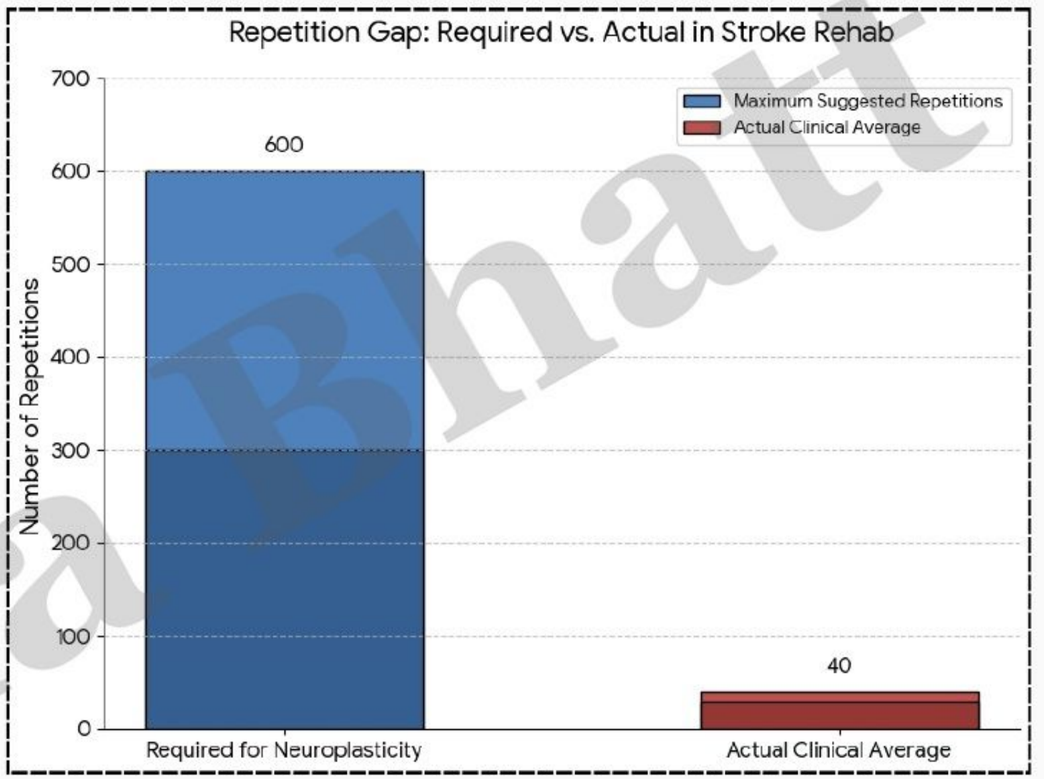
Up to 80% of stroke survivors have upper limb impairement.

Presents prominently during the early acute and sub-acute periods following a stroke, limiting functional independence and complicating early mobilization

50% of survivors demonstrate continued impairement at 6 months

40% of these individuals exhibit "BCI illiteracy," demonstrating too little initial mu-wave EEG modulation without alternative reinforcement.

Sources: AHA/ASA Stroke (2022) Timing & Dose systematic review; JMIR 2022;24(6):e34307 — upper-limb impairment prevalence.



Sources: Counting Repetitions: An Observational Study of Outpatient Therapy for People With Hemiparesis" (Lang et al., 2009).

NeuroReach brings typically resource-intensive pattern into an accessible game any clinic can run on a single PC with consumer-grade hardware.

Krishna Bhatt

NeuroReach's Solution

Designed to augment traditional therapy

< 10 Minutes to complete calibration

1 Computer and lightweight architecture

200+ Attempts for rich session data

Max. Inclusion across all patients

NeuroReach bridges the clinical intensity gap and maximizes high-quality outcomes with limited time.

Severity-Aware Personalization

Difficulty and reward thresholds are dynamically calibrated to each patient's severity, preventing frustration.

Grounded Therapy Adjunct

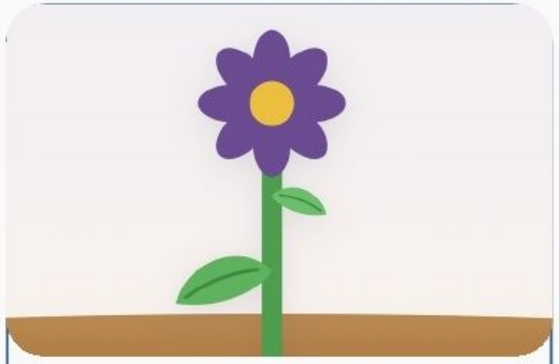
Built to act as a supervised asset alongside standard care rehabilitation as a wellness system.

Evidence-Based Proxy Tracking

Logs session-level proxy metrics (laterality index, ROM, etc.) alongside FMA-UE and ARAT assessments.

Prachna Bhatt

The Patient Experience



Early Recovery

Driven by sustained focus with a non-punitive design.

[Garden Grow]

Late Recovery

Targets mapped to specific ROM goals.

[Star Catcher]



- **Automated Hemiparetic Protocols:** Automatically mirrors tracking landmarks based on the patient's affected side .
- **Multi-Stage Scaling:** Seamlessly transitions across 3 clinical modes as physical movement returns.

And more to come...

- **"Lighthouse Keeper" Storymode**
- **River Rowing**
- **Fruit Picking**

Mode A
100% EEG-driven for severe impairments

Mode B
Scores neural intent and physical effort together

Mode C
Focus to physical range of motion and neural coupling

Stroke-optimized design featuring neglect-aware visual layouts, an icon-first UI for aphasia, and zero-strobe rendering to fully eliminate seizure risks.

Krishna Bhatt

Clinical Workflow





Standard Clinic Slot Structure

Tailored to a standard 30-40 minute appointment window: 6-min calibration, two 9-min gameplay blocks split by rest time.

Automated Progress Reports

Tracks long-term trends, generating clean summaries for internal charts and progression analysis

Cortical Engagement **Postural Adjustements**

Motor Quality **FMA-UE / ARAT Baselines**

Classifier Performance **Range of Motion**

Krishna Bhatt

How it Works

Grounded in established BCI-rehabilitation science

EEG Data Stream
Brain Engagement

Camera Feed
Movement Quality

Intent Score
Driven by Motor Imagery

Therepeutic Feedback
Gamified Neuroplasticity

Full Spec @ https://github.com/blizzard-labs/thalis-rehab-game

EEG-Based Technology

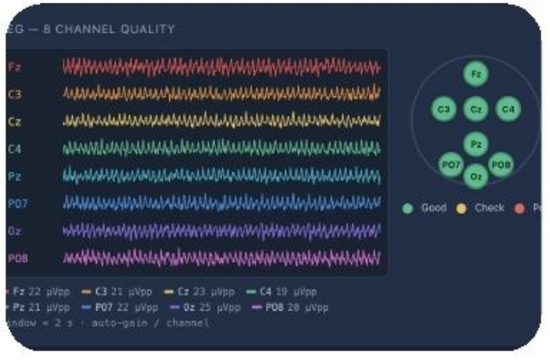
Using a consumer-grade EEG headset (**g.tec Unicorn Hybrid Black**), NeuroReach employs THALis-Micro decoding technology with minimal calibration, monitoring **event-related desynchronization (ERD)**.

Computer Vision Validation

A local webcam tracks skeletal pose and hand keypoints via MediaPipe, validating **movement-free MI trials** while measuring ROM and **trunk-lean compensation**.

A Gamified Experience

The Fusion Engine combines 8-channel EEG and computer vision through a synchronized LSL pipeline, producing a unified **Intent Score with <250 ms end-to-end latency**.





Krishna Bhatt

Safety, Privacy, and Regulatory



- **Transparent Design:** Open-source architecture with full engineering specifications for complete software verification
- **Flexible Deployment:** 100% offline with customizable treatment and privacy protocols per clinic needs.

No Patient Data Leaves the Clinic

Entire processing engine executes locally on the clinic PC and internent LAN. No patient identifiers (PHI) or raw webcam streams ever cross public internet.

Seizure & Electrical Safety

Utilizes a CE-marked, battery-powered EEG headset that is electrically isolated from the main grid. Strict ban on high-contrast flicker or strobing.

IRB-Compliant Architecture

De-identified data and informed consent/assent logging frameworks are built-in, providing an instant turn-key framework for academic clinical pilots.

Krishna Bhatt

Pilot Proposal

NeuroReach is an engineering prototype built from established rehabilitation principles and is now seeking its first supervised clinical pilot.

What I provide	Clinic provides
✓ Equipment	✓ Space
✓ Setup	✓ Therapist supervision
✓ Technical support	✓ Volunteer participants
✓ Data summaries	

- Pilot with 3–10 volunteer patients
- Clinician supervision at all times
- No disruption to standard therapy
- Sessions incorporated into existing appointments
- Feedback used to improve the system

Learn

Evaluate usability.

Measure

Collect therapist feedback.

Improve

Refine games and workflow

No Commitment: Clinic may discontinue participation at any time. No additional time required beyond therapy

Krisha Bhatt